



APEX RETAIL BANK

Customer 360 & Silent Churn Intelligence

Enterprise Data Governance, Forensic Data Quality & Prescriptive Attrition Prevention

Wells Fargo MBA Case Competition | Complete 4-Deliverable Synthesis

Executive Summary: Diagnosing Silent Attrition

[OBSERVED]

WHAT IS HAPPENING?

23.5% of Customer Base (2,402 Customers) exhibit severe silent churn signals—quietly transferring funds out while digital app engagement deteriorates.

[DERIVED]

WHY DOES IT MATTER?

₹67.33 Crores in retail deposits (51.4% of total portfolio) are directly exposed to flight risk, concentrated heavily in high-margin Wealth & Privileged tiers.

[DERIVED]

WHICH SEGMENTS ARE EXPOSED?

Wealth segment accounts for 59.0% of at-risk deposits (₹39.73 Cr) with a 61.0% segment balance risk rate. Top 5 branches hold ₹14.39 Cr at risk.

[RECOMMENDED]

WHAT SHOULD MANAGEMENT DO?

Execute a 6-Archetype Action Plan: prioritized 48-hour RM outreach for high-value clients, KYC & data cleanup, and proactive resolution of dispute-heavy channels.

Deliverable 1: Dataset Discovery & Relational Join Architecture

INVENTORY & GRAIN

- Customer_Master: 10,200 rows | Grain: 1 row/customer | PK: Customer_ID | Missing: 20 PAN, 306 KYC.
- Accounts: 14,000 rows | Grain: 1 row/account | PK: Account_ID | FK: Customer_ID → Customer_Master.
- Transactions: 150,000 rows | Grain: 1 row/txn | PK: Txn_ID | FK: Account_ID → Accounts.
- Loans: 5,000 rows | Grain: 1 row/loan facility | PK: Loan_ID | FK: Customer_ID →

JOIN RISKS & ORPHAN RECORDS

- Orphan Transactions (DQ-022) [OBSERVED]: 1,488 txns reference non-existent Account_IDs. Must be quarantined to avoid phantom postings.
- Orphan Loans (DQ-023) [OBSERVED]: 25 loans reference Customer_IDs not in Customer_Master. Direct credit exposure without KYC backing.
- Join Inflation Prevention [DERIVED]: Direct 1:N fan-out joins between Accounts and Transactions cause massive

MATCHING RULES & ENFORCEMENT

- Customer Anchor: Customer_Master serves as the single anchor entity.
- Deduplication Matching: Customer_ID deduplicated; RapidFuzz multi-key matching across Name + DOB + PAN resolves 2,039 fuzzy candidate pairs.
- Directionality: Strict 1-to-many single-direction filtering enforced.
- Uniqueness Verification: Customer 360 grain verified as

Deliverable 1: 18 Commercial & Analytical Questions Formulated

Commercial Inquiries Connecting Data to Profitability, Liquidity, Service & Risk

1. Customer_Master: Q1 (Compliance) Segment KYC failure rates? | Q2 (Governance) Duplicate clusters sharing PAN/contact? | Q3 (Operations) Data quality profiles across onboarding cohorts.
2. Accounts: Q4 (Liquidity/Risk) Branch balance concentration risk? | Q5 (Profitability) Multi-account customer profitability vs single-account? | Q6 (Service) RM portfolio value overload and capacity limits.
3. Transactions: Q7 (Profitability) Net money movement (credit minus debit) by segment? | Q8 (Churn) Channels & merchant categories dominating high-value outflows? | Q9 (Velocity) Trailing 90D debit acceleration.
4. Loans: Q10 (Credit Risk) NPA concentration across loan products? | Q11 (Cross-Signal) Correlation between high DPD and service complaints? | Q12 (Stressed Assets) Total exposure in 60+ DPD status by segment.
5. Customer_Service: Q13 (Service Quality) Categories with worst resolution TAT & lowest CSAT? | Q14 (Churn) Linkage between fee disputes and large fund transfers? | Q15 (Operations) Branch complaint skew.

Deliverable 2: Forensic Data Quality Scorecard & Remediation

7 CRITICAL DEFECTS

- Missing PAN (DQ-001): 20 accounts missing tax identifiers. Remediate via tax portal.
- Duplicate PAN (DQ-007): 480 accounts sharing PAN credentials. Major AML/tax breach.
- Invalid PAN Regex (DQ-009): 82 format violations. Implement gateway regex check.
- Orphan Transactions (DQ-022): 1,488 txns tied to non-

7 HIGH-SEVERITY DEFECTS

- Missing KYC Status (DQ-004): 306 customer records incomplete.
- Duplicate Emails (DQ-008): 352 accounts sharing emails.
- Invalid Email Format (DQ-010): 102 syntax errors.
- Future Onboarding Dates (DQ-024): 46 temporal anomalies.
- Future Account Open Dates (DQ-025): 66 accounts opened up to 2027

REMEDIATION & CONTROLS

- Physical Quarantine: Violating records isolated in data/quarantine/ (13 CSV extracts).
- Entity Resolution: RapidFuzz matched 2,039 fuzzy duplicate pairs in entity_resolution.xlsx.
- Preventive Gateway: Enforce schema-level constraints on PAN regex and future dates.
- Detective Automation: Automated daily DQ script alerts ExCo to newly surfaced anomalies

Deliverable 3: Enterprise Business Glossary & Data Dictionary

CUSTOMER_MASTER (9 FIELDS)

- Customer_ID: VARCHAR | Pattern: ^CUST_\d{5}\$ | High Trust (System PK) | Steward: Core Banking Ops.
- Name: VARCHAR | Free text | Title cased, Unicode NFKD clean | Steward: Onboarding Desk.
- DOB: DATE | ISO 8601 | Range: 18–100 yrs | High Trust (KYC verified) | Steward: Compliance.
- PAN: CHAR(10) | Regex: ^[A-Z]{5}[0-9]{4}[A-Z]\$ | Medium

LOANS & CREDIT (4 FIELDS)

- Loan_ID: VARCHAR | Pattern: ^LN_\d{5}\$ | Unique credit facility identifier | Steward: Lending Ops.
- Loan_Amount: DECIMAL(14,2) | Sanctioned principal | Range: > 0 | Steward: Credit Underwriting.
- DPD_Days: INTEGER | Days past due | Range: 0–360+ | Primary delinquency metric | Steward: Collections.
- NPA_Flag: CHAR(1) | Permitted: 'Y' 'N' | RBI IRAC

ACCOUNTS, TXN, CS & DIGITAL

- Balance: Ledger deposit balance in INR | Crucial for liquidity & churn impact | Steward: Treasury.
- Amount: Transaction debit/credit magnitude in INR | Steward: Payment Operations.
- CSAT_Score: Integer scale (1–5) | Post-resolution satisfaction | Steward: Head of CX.
- Resolution_TAT_Days: Days to close grievance | SLA monitoring | Steward: Service

Deliverable 4: Customer 360 & Silent Churn Scoring Model

Mathematical Formulation of the Silent Churn Risk Index (0–100 Scale)

- Silent Churn Risk Index = Dim 1 (Customer Value) + Dim 2 (Outflow Signal) + Dim 3 (Digital Decay) + Dim 4 (Service Friction) + Dim 5 (Credit Stress) + Dim 6 (Data Confidence Adjustment).
- Dim 1 — Customer Value (0–20 pts): Balances (40%), Account diversity (15%), Loan relationship (15%), Segment tier (30%). Measures business impact if customer leaves.
- Dim 2 — Outflow Signal (0–25 pts): Recent debit volume (40%), Trailing 90D debit acceleration ratio (30%), Transaction count drop (30%). Detects active capital flight.
- Dim 3 — Digital Deterioration (0–20 pts): Login recency (35%), Session frequency (35%), Digital engagement trend (30%). Identifies digital disengagement 60 days before closure.
- Dim 4 — Service Friction (0–20 pts): Complaint count (25%), Recent complaints (25%), Inverted CSAT score (30%), Resolution TAT (20%). Emotional catalyst for churn.
- Dim 5 — Credit Stress (0–10 pts): Max DPD (60%), NPA flag (40%). Financial distress deteriorating

Deliverable 4: Portfolio Exposure — Customer Value vs. Risk



Segment Exposure Matrix [DERIVED]

- Wealth Tier: 865 Customers (8.5% of base) | Total Deposits: ₹65.13 Cr | High Risk: 463 Customers (53.5%) | Balance at Risk: ₹39.73 Cr (61.0% of segment). Severe capital concentration.
- Privileged Tier: 2,099 Customers (20.6% of base) | Total Deposits: ₹37.83 Cr | High Risk: 824 Customers (39.3%) | Balance at Risk: ₹19.67 Cr (52.0% of segment). High digital banking sensitivity.
- Mass Retail Tier: 7,236 Customers (70.9% of base) | Total Deposits: ₹28.12 Cr | High Risk: 1,115 Customers (15.4%) | Balance at Risk: ₹7.94 Cr (28.2% of segment). Service friction & fee dispute drivers.
- Strategic Takeaway [RECOMMENDED]: Retention efforts must not be democratic. Wealth accounts carry ₹39.73 Cr in at-risk balances (5x the entire Mass Retail exposure). Immediate frontline focus on top 463 at-

Deliverable 4: Triangulated Early Warning Churn Indicators

Dimension 2 (Weight: 25%)

MONEY OUTFLOW SIGNALS

- Outflow Volume: 2,599 customers show recent debit spikes > 75th percentile.
- Outflow Velocity: Ratio of 90D debit to historical debit accelerating by > 2.0x.
- Transfer Activity: Rapid UPI/NetBanking transfers to external institutions without commensurate inward credit.

Dimension 3 (Weight: 20%)

DIGITAL DETERIORATION

- Login Inactivity: 3,781 customers exhibit significant drop in session recency (> 45 days since login).
- Feature Abandonment: App usage drops from transaction features to purely checking balances.
- Session Duration: 35% decline in monthly app time preceding primary fund transfers.

Dimension 4 (Weight: 20%)

SERVICE FRICTION SIGNALS

- Grievance Clustering: 1,480 customers logged multiple grievances within 90 days.
- CSAT Collapse: Severe churn correlation among customers giving CSAT ≤ 2 .
- Resolution Delay: Outlier resolution TAT (> 20 days) creates a 3.4x higher churn propensity.

Deliverable 4: Explainable Segmentation — 6 Actionable Archetypes

Actionable Frontline Archetypes & Prescriptive Operating Protocols [RECOMMENDED]

- Archetype A — High Value + Multi-Signal Risk (165 Customers | ₹23.62 Cr): High deposit balance + simultaneous outflow, digital decay, and complaints. → PROTOCOL: Priority Senior RM outreach within 48h; personalized wealth retention offer.
- Archetype B — High Outflow + Weak Supporting Evidence (599 Customers | ₹7.90 Cr): Large single outflow without app decline or complaints (e.g., tax payment or real estate purchase). → PROTOCOL: 30-day monitoring; DO NOT harass customer with churn retention calls.
- Archetype C — Digital Decline + Service Friction (2,779 Customers | ₹27.27 Cr): Frustrated digital users with low CSAT and declining logins. → PROTOCOL: Digital service recovery call, fee reversal where applicable, proactive app re-onboarding.
- Archetype D — Credit Stress Driven Risk (1,193 Customers | ₹9.96 Cr): Elevated loan DPD (30–90 days) impacting broader banking relationship. → PROTOCOL: Branch credit counseling, debt restructuring review, proactive restructuring counseling.

Deliverable 4: Service Friction & Digital Abandonment Dynamics

GRIEVANCE DRIVERS

- Total Complaints: 12,000 across 7,077 customers.
- Top Category: Transaction Disputes (34%) & Mobile Banking Failures (28%).
- Worst Channel: Branch Queue complaints yield the lowest CSAT (average 2.1 / 5.0).
- TAT Outliers: 412 complaints took > 30 days to resolve (standard SLA: 7 days).

DIGITAL APP USAGE

- Total Activity: 150,000 sessions across 8 app modules.
- Critical Drop-off: 'Transfers' and 'KYC' pages account for 58% of abandoned sessions.
- Silent Churn Leading Indicator: When monthly sessions drop below 2, outflow probability doubles within 60 days.
- Inactivity Clustering: 22% of high-risk customers have zero logins in trailing 60 days.

THE ATTRITION FLYWHEEL

1. Digital Friction: App transfer failure or login error.
2. Service Delay: Customer raises complaint; TAT exceeds 14 days.
3. Trust Erosion: Customer rates CSAT 1 or 2.
4. Silent Outflow: Customer migrates primary deposits to competitor bank.
5. Account Becomes Dormant: Bank loses customer without notice.

Deliverable 4: Regional Governance & Branch Risk Concentration

Top 5 Branch Hotspots Account for ₹14.39 Cr of At-Risk Deposits [DERIVED]

- Branch BR004 (Metropolitan Commercial Hub): 243 Customers | Total Deposits: ₹4.68 Cr | Balance at Risk: ₹3.26 Cr (69.5% risk rate) | 281 complaints, CSAT 3.75. Driven by Wealth tier outflow and digital friction.
- Branch BR001 (Urban Flagship Center): 285 Customers | Total Deposits: ₹4.36 Cr | Balance at Risk: ₹3.17 Cr (72.8% risk rate) | 350 complaints, CSAT 3.58, 2 NPAs. High service friction and dispute concentration.
- Branch BR023: 196 Customers | Total Deposits: ₹4.50 Cr | Balance at Risk: ₹2.82 Cr (62.6% risk rate) | 238 complaints. High average customer balance with declining mobile engagement.
- Branch BR014: 254 Customers | Total Deposits: ₹3.89 Cr | Balance at Risk: ₹2.60 Cr (66.7% risk rate) | 310 complaints. High debit outflow acceleration to competitor accounts.
- Branch BR012: 265 Customers | Total Deposits: ₹4.08 Cr | Balance at Risk: ₹2.54 Cr (62.2% risk rate) | 322 complaints, 2 NPAs. Dual service delays and credit delinquency.

Prescriptive Management Action Matrix & Governance Controls

Operational Accountability Matrix & 3-Tier Governance [RECOMMENDED]

- 1. High-Value Multi-Signal Flight → Priority Outreach: Personal call by Senior RM within 48 hours; bespoke deposit rate / wealth fee waiver. | Owner: Head of Wealth Banking | Priority: P1 - Critical.
- 2. Service Disruption Trigger → Service Recovery Protocol: Automatic grievance escalation for high-balance clients; discretionary fee reversal up to ₹5,000 for verified delays. | Owner: Head of Customer Experience | Priority: P1 - Immediate.
- 3. Digital Inactivity Warning → Re-engagement Nudge: Automated SMS/email prompt highlighting upgraded app features and biometric quick-login. | Owner: Head of Digital Channels | Priority: P2 - High.
- 4. Credit Stress Cluster → Proactive Restructuring: Outreach by branch credit officers before 90-day DPD cliff; loan tenure extension options. | Owner: Chief Risk Officer / Retail Credit | Priority: P2 - High.
- 5. Identity & KYC Defects → Digital Document Portal: In-app video-KYC link sent to 480 duplicate PAN and 306 missing KYC accounts. | Owner: Head of Compliance & Operations | Priority: P1 - Regulatory.

6. Regulatory & Data Privacy Compliance: Quarterly API audits for data integrity / consent management. | Owner: Head of Compliance & Operations | Priority: P2 - High.

Strategic Execution Roadmap: 30-60-90 Day Phasing

DAYS 1–30: TRIAGE & HIGH-VALUE RETENTION

- Deploy RM Outreach Console to all Wealth Relationship Managers.
- Contact Top 165 Archetype A customers (protect ₹23.62 Cr).
- Remediate 480 duplicate PAN identities and 306 missing KYC records.
- Establish weekly DQ exception monitoring meetings.

DAYS 31–60: OPERATIONAL REPAIR & BRANCH INTERVENTION

- Dispatch service recovery squads to Top 5 risk branches (BR004, BR001, BR023, BR014, BR012).
- Implement 7-day SLA cap on Transaction Dispute complaint categories.
- Launch mobile app re-engagement push for 3,781 inactive digital accounts.
- Ingest daily Power BI dashboards for regional directors.

DAYS 61–90: INSTITUTIONALIZATION & AUTOMATION

- Automate daily lakehouse ingestion pipeline from core banking systems.
- Integrate Silent Churn Risk Index into frontline CRM screen (real-time risk badge).
- Calibrate credit stress early-warning triggers with loan collection units.
- Measure net deposit retention: Target ₹30+ Cr in preserved retail deposits.